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BIH Competence Tracker

Week 1 Research Brief: Data & Domain Modelling

Audience	Data Engineer & Backend Developer
Research Period	Week 1
Primary Focus	Relational database design, domain modelling, PostgreSQL, data integrity, and preparation for Django implementation

PURPOSE

The purpose of Week 1 is to establish a strong **data and domain foundation** for the BIH Competence Tracker before significant application development begins.

The **BIH Competence Tracker** will serve as the official platform for **participant registration, learning-path enrollment, level progression, competence tracking, assessment grading, assessor feedback, GitHub submission references, final BIH grades, progression decisions, and participant completion records.**

GitHub will remain the **primary environment** for **code submissions**, projects, pull requests, automated testing, technical evidence, and code review.

Key research question:

How should BIH represent **participants, programmes, levels, competencies, assessments, evidence, results and progression** in a relational database while maintaining data integrity, configurability and historical records?

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The expected result is not a finished production database. The outcome should be a researched, documented and defensible data model that can form the basis of the first **Django implementation.**

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1. THE DOMAIN MODEL (MR S.G. MBHAMALI)

1.1. Should BIH use **Django's authentication** user as the central identity?

- The best way to handle this is to use a **Custom User Model** as the central identity, and then use Django's built-in flags to separate who is an admin and who is a participant. But then using the Custom user model does not mean we are out of the use of **Django's authentication user**.
- 1. **Admins (Us):** Django already has a built-in flag called `is_staff`. When this is set to `True`, that user can log into the Django Admin panel to manage courses, grade work, and issue certificates.
- 2. **Participants (Students):** Anyone where `is_staff` is `False` is treated as a regular participant. They will only see the main student website where they study and take tests.
 - **Participant Entity:** A user where `is_staff = False`.
 - **Admin Entity:** A user where `is_staff = True`.

1.2. What information belongs on User versus ParticipantProfile?

- According to our software we have two **user** entities: **Participant Entity & Admin Entity**.
- **ParticipantProfile** Belongs under the Entity called **Participant Entity**. The Participant profile contain the participant details or data, their full name, profile picture, etc etc...

1.3. Can one user hold multiple roles? (refer to Babongile's document on Access Control)

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- Yes, multi-roles can be done, one user can use the site as an admin and participant.

1.4. Could an assessor also be a participant?

- Yes, for Testing if assessor want to experience the participant side, and see if other features are working correctly.

1.5. Should roles use Django Groups/Permissions, profile models, explicit role fields, or combination?

- For BIH, the absolute best approach is a **combination** of an explicit role field and Django Groups/Permissions.

1.6. What information must remain available when a participant becomes inactive?

- When a participant becomes inactive in BIH, you should **never delete their account**. Instead, you should simply change their status flag to inactive (`is_active = False`).
- Because BIH is a platform that handles **grading and certifications**, certain records must stay in your database forever. This ensures the integrity of your system and protects both the student and your organization.



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2. LEARNING PATHS & LEVELS (MR ANN SIBISI)

2.1 Independency of Learning Paths Levels

A level in Beacon Innovation Hub context is like a stage that is defined by assessments and learning objectives that a user or a participant must master before they move to the next stage for each learning path. Instead of defining each level as vague words like 'Beginner' or 'Advanced', we can best define them as **structured competency**.

Example:

Backend Development

- **Level 1 — Foundations:** Version Control (Git/GitHub), Command Line, Programming Fundamentals (Python/Node.js).

Frontend Development

- **Level 1 — Web Core:** HTML5, CSS3, Responsive Layouts, Basic JavaScript

This type of level description makes each level definition unique for each learning path.

2.2 Can the same level definition be reused for all learning paths?

The structure for defining each level can be used for any level definition in all learning paths. All levels maintain the same definition structure since each level must define the requirements that must be **completed prior to entry, specific skills to be evaluated** and the **final task that must be completed to advance into another level**.

2.3 How must levels be ordered?

The levels must be ordered in such a way that each level advances from the previous one/s. This ensures that all the requirements for each level are completed in the previous levels before the next level is attempted.

2.4 Multi-Enrollment

Based on the Hub's policy the participant cannot enroll in more than one learning path, as this will distort the work flow of the team and the participant.

2.5 Withdrawal, Suspension, and Completion

This defines the participant's status of enrollment to the course. The participant's status will depend on how the participant progresses with the course.

Withdrawal

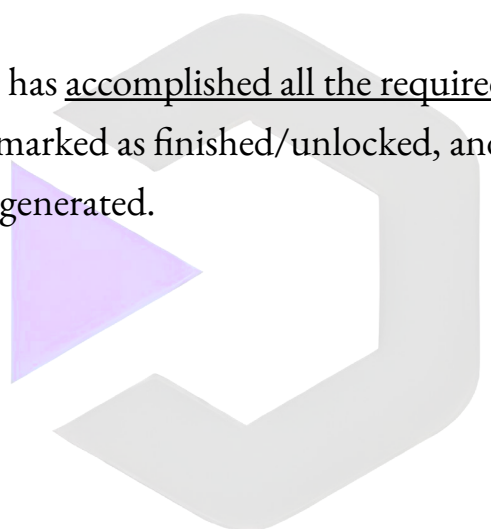
This refers to the participant leaving the course midway without completing it. Many factors may contribute to such results which include personal reasons as well. The participant's status will reflect WITHDRAWN.

Suspension

This will be the result of not complying with the community policies such as: [late submissions](#) without any valid reason or breaking the [AI plagiarism or any other form of other plagiarism policy](#) of the Hub. The participant's profile will be flagged and the participant will not be able to continue no longer.

Completion

This means the participant has accomplished all the required assessments and all the competences. All levels are marked as finished/unlocked, and official certificates or completion records can be generated.



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3. COMPETENCY (MR P.G MARAPIRA DHLIWAYO)

3.1. What is competency?

In the **Beacon Innovation Hub's** context, competency refers to having a skill that can be applied to achieve a certain goal effectively.

When an individual is considered competent at **BIH**, they are able to use their skill to execute any given task. The point is not having the skill, but it is the ability to apply the skill.

3.2. Competency primarily includes:

- Knowledge of **tools** e.g. Python Programming
- Knowledge area such as **HTTP**
- A **behaviour** such as professional coding practices

3.3. Competencies should be granted for:

- **Knowledge and application** of tools e.g. Python Programming
- Completion by displaying knowledge in a certain **LEVEL** on a PATH.
e.g. A junior Data Scientist competency can be granted to a participant who has completed **LEVEL 1 - Foundational Data Science**
- Professional Application of skills. e.g. A participant has completed all the levels of Data Scientist & Machine learning incl. all professional working standards

3.4. Can competencies be shared?

No, in the BIH context each learning path e.g. Data Analytics has its own levels that may be independent of each other. Hence there will not be any shared competencies.

3.5. How competence is demonstrated?

The BIH level competence is demonstrated by considerable evidence on a GitHub repository that will be tested for its **code quality, security and functionality**. This is supplemented with **Technical Defense** of architecture (via a Teams Meeting) and **making live code changes** to demonstrate potential/possible changes.

3.6. Competence Grading

To **grade the BIH competence**, three main factors must be considered to give an output score for the final review. Namely:

❖ **GitHub Evidence** - The code presented on GitHub must be correctly functional and able to give the expected OUTPUT, the **code quality** must be considered including coding standards. **(60%)**

❖ **Technical Defense** - The evidence presented on the GitHub repository must then be used to defend all the architecture used via a Teams meeting **(30%)**

❖ **Live modifications & Documentation** - The assessors will ask for multiple modifications to be made on the code live and test software scalability, portability & maintenance **(10%)**

(To qualify as a competent participant, one must obtain a grade of **65%+** as a combined score of the three criteria)

3.7. Storage of competence results

The competence scores will be stored in the format of **(0 - 100%)**

Combining all the different criteria mentioned, this score must be associated with a **review** such as (Not Assessed, Assessment In Progress, Competent, Not Yet Competent, Reassessment Required, Reassessment Scheduled ,Finalized) and feedback from the assessor.

3.8. Reassessment

If a candidate does not meet the **65%** criterion, they do not qualify for the competence certificate but they can take a **reassessment** where they will be tasked with another competence assessment. The **old assessment** will be used as guided practice to be able to use analytical skills, critical thinking etc., to be able to solve problems.

The difficult scenario: (**AI** generated):

Thando is enrolled in Backend Development. At Level 2, Django Development, she **submits one GitHub project**. The project assesses Python Programming, Git/GitHub, Database Design, Django and Backend Security. She performs well in Python and Git but poorly in Django and Backend Security. Her overall assessment grade is 76%, which is above the level's pass mark.

- ☐ Is a cumulative **76% score** sufficient for a **competent** status, or does the Hub require a **separate validation** of specific skills?

Thando will not be able to even get the grade of **76%** because, the final score for the competence is accumulated by testing out all competencies e.g. the final score must include Python, Git/GitHub, Database design, Django & backend security.

- ☐ Does the Hub permit **Thando's advancement** despite her failure to meet the **competency benchmarks** in essential technical areas?

No, Thando cannot advance without displaying actual competence for her skills and fields/areas of specialization since the tasks on the lower end are all foundational and prerequisite to be able to advance to other stages.

- ☐ To ensure accountability in the future, what **historical records** must the Hub maintain to justify her **progression decision** after six months have passed?

BIH must maintain a **history record** of her scores, reviews and feedback leading to the final decision on whether she continues or not.

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REFERENCES:

1. RESEARCH 1 - COMPETENCE TRACKER (RESEARCH BRIEF)



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